

FLAT: Resampling Image and Text into 1D Flexible-Length Aligned Transmodal Tokens for Retrieval and Generation

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Traditional multimodal representation learning and generation are two stages: a contrastive or self-supervised visual encoder is trained first, followed by a separate downstream generative model. This setup bottlenecks generative performance behind frozen embeddings. To bridge this gap, we revisit joint multimodal representation learning and generation to produce linearly interpolatable embeddings that are directly consumable by generative decoders. We present FLAT (**F**lexible-**L**ength **A**ligned **T**ransmodal representations), a representation pre-training framework that jointly optimizes a shared multimodal encoder alongside downstream text-to-image (T2I) and image-to-text (I2T) decoders. By combining contrastive alignment with bidirectional cross-modal generative objectives, FLAT ensures its representations function as both discriminative semantic descriptors and generative conditions. Architecturally, FLAT maps visual and textual inputs into a unified continuous 1D sequence space, applying nested dropout over prefix- K tokens to enable dynamic output lengths. A single pre-training stage allows FLAT to perform cross-modal retrieval and generation across variable prefix K , achieving a T2I GenEval score of 71.1. Task-specific fine-tuning aligns model performance with state-of-the-art baselines: 83.1 GenEval on T2I generation; 40.5 BLEU-4 and 138.6 CIDEr on MS-COCO image captioning; and Recall@5 scores of 86.8 (I2T) / 75.8 (T2I) on MS-COCO alongside 98.3 (I2T) / 93.6 (T2I) on Flickr30K. Finally, qualitative evaluations demonstrate that FLAT representations natively support linear interpolation, latent space arithmetic, and zero-shot composed retrieval.

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1 Introduction

Traditional multimodal architectures separate modality representation learning from cross-modal generation. In representation learning, models like CLIP (Radford et al., 2021) align visual and textual features using contrastive objectives, or DINO (Caron et al., 2021) and JEPA (Assran et al., 2023) train visual with self-supervision. In image-to-text (I2T) models, the frozen visual representations are connected to a transformer decoder and re-aligned with the text embedding space (Liu et al., 2023). Conversely, text-to-image (T2I) models rely on pre-trained, frozen text encoders: Stable Diffusion and SDXL utilize CLIP text embeddings, SD3 combines CLIP and T5, PixArt and Sana and MetaQuery leverage frozen large language models (Radford et al., 2021; Rombach et al., 2022; Podell et al., 2024; Esser et al., 2024; Chen et al., 2024; Xie et al., 2025; Pan et al., 2025). This decoupling of representation learning and generative modeling bottlenecks generative performance behind frozen representations and requires re-alignment in generative model training.

To bridge this gap, recent unified multimodal models either jointly train representation encoders with transformer backbones using generative objectives, or bypass vision encoders to operate directly in pixel space (Agrawal et al., 2024; Diao et al., 2025; Liu et al., 2026). In this work, we revisit an alternative direction: retaining an explicit contrastive, linearly-interpolatable embedding space while simultaneously optimizing alignment and generation. We present **FLAT** (**F**lexible **L**ength **A**ligned **T**ransmodal representations), a representation pre-training framework that jointly trains a multimodal encoder with T2I and I2T decoders using **dual bidirectional alignment and generative** losses. Our central insight is that retrieval and generation are

mutually reinforcing (Yu et al., 2022)—effective representations should serve as both discriminative semantic descriptors and generative conditions for either direction. We demonstrate that these combined objectives produce representations that are linearly-interpolatable and consistently enhance downstream tasks.

FLAT builds upon recent advances in 1D visual tokenization, which resample image grids into compact 1D token sequences to eliminate spatial redundancy (Yu et al., 2024; Bachmann et al., 2025). Similar techniques have been employed to compress lengthy text into summary vectors (Chevalier et al., 2023). Unlike prior visual tokenizers that focus on image reconstruction, FLAT resamples both visual and textual inputs into a unified 1D representation space via a shared multimodal encoder, producing continuous 1D representations that are contrastively aligned during generative training. FLAT incorporates nested dropout (Bachmann et al., 2025; Rippel et al., 2014; Kusupati et al., 2022) into sequence representation learning by optimizing over random prefix- K tokens per batch, enabling flexible sequence length selection at inference time.

We evaluate FLAT across T2I and I2T generation and retrieval. By varying the number of prefix- K tokens, one pre-trained FLAT model retrieves and synthesizes coarse-to-fine cross-modal outputs from the same encoder pass. After a single pre-training stage it reaches a GenEval score of 71.1 on T2I generation and, zero-shot, MS-COCO Recall@5 of 69.1 (I2T) and 64.6 (T2I). Task-specific fine-tuning then brings model performance on each task in line with published SOTA baselines: 83.1 GenEval on T2I generation, above every baseline we compare against including two 7B models that additionally rewrite the prompt; 40.5 BLEU-4 and 138.6 CIDEr on MS-COCO for I2T generation; and Recall@5 of 86.8 (I2T) / 75.8 (T2I) on MS-COCO and 98.3 (I2T) / 93.6 (T2I) on Flickr30K. On ImageNet linear probing the frozen representation reaches 81.8, above all published results for generative latent space. Finally, qualitative evaluations show that FLAT representations natively support interpolation, semantic arithmetic on latent space, and zero-shot composed retrieval.

2 Related Work

Decoupled Representation Learning and Generation. Contrastive representation learning methods in the CLIP family (Radford et al., 2021; Jia et al., 2021; Zhai et al., 2023; Sun et al., 2023) have established the benchmark for cross-modal alignment and retrieval. However, downstream multimodal generative models typically build on separately pretrained encoders. In I2T architectures such as BLIP-2 (Li et al., 2023b) and LLaVA (Liu et al., 2023), frozen visual encoders are connected to large language models through trainable adapters. Conversely, T2I models such as unCLIP (Ramesh et al., 2022), SDXL (Podell et al., 2024), SD3 (Esser et al., 2024), PixArt- α (Chen et al., 2024), and Sana (Xie et al., 2025) use frozen text or multimodal encoders as conditioning modules. This separation persists in many unified multimodal systems, which keep pretrained visual encoders, tokenizers, or VAEs frozen while training the components for understanding and generation (Wu et al., 2024a; Chameleon Team, 2024; Xie et al., 2024; Zhou et al., 2025; Pan et al., 2025; Chen et al., 2025a).

Joint Representation Learning and Generation. BLIP (Li et al., 2022) and CoCa (Yu et al., 2022) combine contrastive image-text alignment with text generation. MAGE (Li et al., 2023c) combines masked image generation with self-supervised representation learning, while DREAM (Li et al., 2026) jointly optimizes image-text contrastive alignment and masked image generation. Meanwhile, recent unified multimodal models process understanding and generation within a shared architecture using generative training (Huang et al., 2025; Liu et al., 2026; Diao et al., 2026). Another line develops unified visual tokenizers or encoders that support both semantic understanding and visual reconstruction (Wu et al., 2024b; Ma et al., 2025; Qu et al., 2025; Zhao et al., 2025; Lin et al., 2025; Fan et al., 2025; Yue et al., 2026; Zhang et al., 2026). FLAT instead jointly optimizes a variable-length continuous representation for contrastive retrieval and bidirectional cross-modal generation, covering both I2T and T2I.

Resampled 1D Representations. TiTok (Yu et al., 2024), FlexTok (Bachmann et al., 2025) and GigaTok (Xiong et al., 2025) resample spatial image grids into compact 1D token sequences. AutoCompressor (Chevalier et al., 2023) resamples long text inputs into summary vectors. BLIP-2 (Li et al., 2023a) resamples image encoder outputs with learnable queries for image-text alignment. MiniGPT-5 (Zheng et al., 2023), DreamLLM (Dong et al., 2024), and MetaQuery (Pan et al., 2025) map or query multimodal LLM hidden states to condition image generation. In comparison, FLAT resamples both images and text into a shared continuous space as

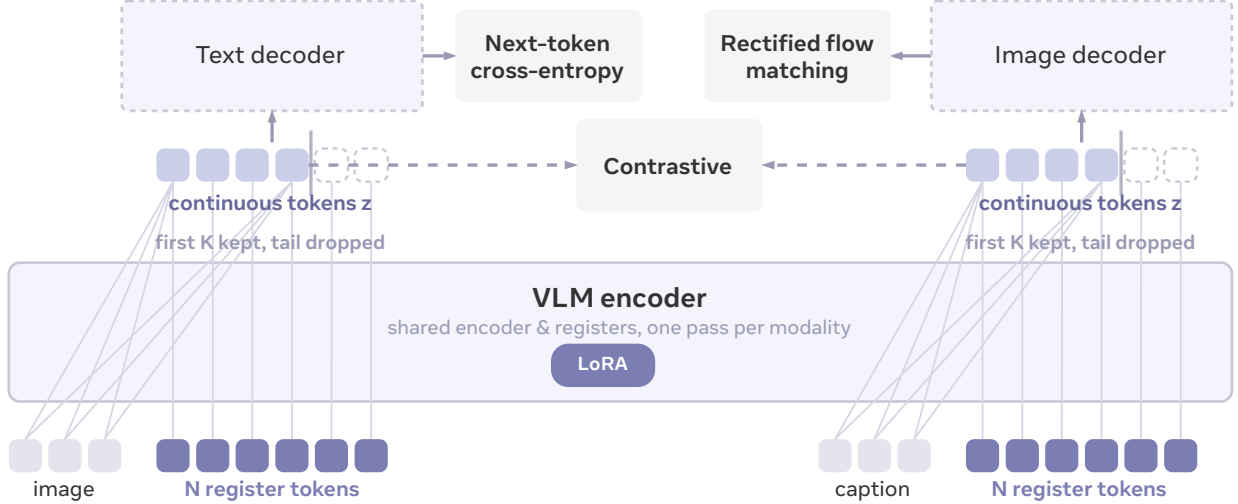


Figure 1 FLAT architecture overview. A shared VLM encoder processes visual and textual inputs through register tokens, applying nested dropout to truncate sequences to variable prefix lengths during joint training. FLAT’s unified 1D continuous representation simultaneously drives contrastive alignment, autoregressive caption generation, and rectified-flow image synthesis.

1D sequence representations.

3 Method

3.1 Overview

FLAT is a representation pre-training framework which learns a multimodal encoder and two transmodal decoders jointly with contrastive and generative objectives. Given an input text or image, FLAT maps the input to an ordered sequence of continuous tokens, which we call the *representation*. The representation is contrastively aligned and read by both transmodal decoders. An overview of FLAT architecture is shown in Fig. 1.

Representation Encoder. let x denote an input of modality $m \in \{\text{img}, \text{txt}\}$. FLAT appends N learnable register tokens $R = \{r_n\}_{n=1}^N$ to each input and encode the resulting sequence with a shared multimodal encoder f_{enc} , which is a pre-trained Vision-Language Model (VLM) with a system prompt *Represent the input*. The hidden states at the register positions are linearly projected to obtain the representation $\mathbf{z}$:

$$\mathbf{z} = W_{\text{lat}} f_{\text{enc}}(x, R) \in \mathbb{R}^{N \times d}, \quad (1)$$

where d is the register dimension and W_{lat} the projection. The two modalities share the same registers and the same projection.

Representation Dropout. To enable variable sequence lengths, we apply nested dropout (Rippel et al., 2014) directly over the representation $\mathbf{z}$. For each training batch, a keep-length K is sampled from a geometric ladder $\mathcal{K} = \{1, 4, 16, 64, 256\}$, and only the prefix sequence $\mathbf{z}_{:K}$ is passed to the contrastive objective and downstream decoders. Sampling from a geometric ladder rather than continuous integers avoids wasting optimization steps on imperceptible length differences. By default, FLAT samples uniformly from $\mathcal{K}$. Appendix F.3 compares uniform sampling with other sampling strategies.

Compared to FlexTok (Bachmann et al., 2025), which maintains a fixed sequence length by padding tail positions with learned null embeddings, we find that directly truncating the representation to the prefix $\mathbf{z}_{:K}$ yields better results in FLAT training setups (Appendix I). To ensure consistency, the target length K is sampled once per training step and broadcast globally across all ranks. This guarantees that both contrastive and generative objectives operate on the exact same sequence prefix at every optimization step.

Decoders. The truncated representation sequence $\mathbf{z}_{:K}$ serves as the semantic condition for both the I2T and T2I decoders. Given a paired image-text input, the visual representation $\mathbf{z}_{:K}^{\text{img}}$ conditions caption generation, while the textual representation $\mathbf{z}_{:K}^{\text{txt}}$ conditions image synthesis. For each generative pathway, $\mathbf{z}_{:K}$ is treated as a sequence of soft tokens and mapped into the respective decoder’s input dimension h :

$$\mathbf{e} = \text{RMSNorm}(\text{MLP}(\mathbf{z}_{:K})); \in \mathbb{R}^{K \times h}, \quad (2)$$

where the MLP projections consist of decoupled, task-specific parameters for the I2T and T2I decoders.

The I2T decoder is a standard autoregressive language model. The visual soft tokens $\mathbf{e}^{\text{img}}$ are prepended with a standard system prompt ("*Describe the input*") and fed as input tokens to autoregressively generate the corresponding text caption. The T2I decoder is a standard rectified flow transformer that denoises continuous VAE latents. The textual soft tokens $\mathbf{e}^{\text{txt}}$ act as semantic conditioning signals integrated via cross-attention mechanisms, guiding the flow-matching model to synthesize VAE latents decodable into the target image. The detailed, multi-task training objectives governing these decoders are formally defined in the next section.

3.2 Joint Training

Given a paired input text-image, FLAT is trained with both bidirectional contrastive loss and generation loss. Since soft tokens $\mathbf{e}$ is projected from representation $\mathbf{z}$, we use $\mathbf{z}$ consistently as conditions in loss functions.

Contrastive Loss. We contrast corresponding register positions in the representation $\mathbf{z}_{:K}$. For an image-text pair (i, j) we define the similarity computed over their total K register positions as a late-interaction score between registers at matching positions:

$$s_{ij}^{(K)} = \frac{1}{K} \sum_{n=1}^K \frac{(z_{i,n}^{\text{img}})^\top z_{j,n}^{\text{txt}}}{\|z_{i,n}^{\text{img}}\|_2 \|z_{j,n}^{\text{txt}}\|_2}, \quad (3)$$

and the contrastive loss $\mathcal{L}_{\text{align}}^{(K)}$ as

$$\mathcal{L}_{\text{i2t}}^{(K)} = -\log \frac{\exp(s_{i,i}^{(K)})}{\sum_{j=1}^{B_{\text{glob}}} \exp(s_{i,j}^{(K)})}, \quad \mathcal{L}_{\text{t2i}}^{(K)} = -\log \frac{\exp(s_{i,i}^{(K)})}{\sum_{j=1}^{B_{\text{glob}}} \exp(s_{j,i}^{(K)})}, \quad (4)$$

$$\mathcal{L}_{\text{align}} = \frac{1}{2} (\mathcal{L}_{\text{i2t}}^{(K)} + \mathcal{L}_{\text{t2i}}^{(K)}). \quad (5)$$

where B_{glob} denotes the size of the gathered global candidates from all ranks.

Generation Loss. For the same pair of image-text, we apply symmetrical generation losses for I2T and T2I respectively, both using the representations $\mathbf{z}_{:K}$ as semantic conditions.

For I2T generation and an image input with representation $\mathbf{z}_{:K}^{\text{img}}$, the loss over target caption y and decoder parameter θ is defined as:

$$\mathcal{L}_{\text{txt}} = -\sum_t \log p_\theta(y_t | y_{<t}, \mathbf{z}_{:K}^{\text{img}}) \quad (6)$$

For T2I generation and a text input, the representation $\mathbf{z}_{:K}^{\text{txt}}$ are used as semantic conditions for a flow-matching decoder. Let $\hat{x}$ be the clean VAE image latent, $\epsilon \sim \mathcal{N}(0, I)$, and $t \in [0, 1]$. The noised VAE latent and target velocity used in flow matching are defined as

$$\hat{x}_t = (1-t)\hat{x} + t\epsilon, \quad v^* = \epsilon - \hat{x}. \quad (7)$$

The conditional flow-matching objective over decoder parameter ϕ is defined as:

$$\mathcal{L}_{\text{img}} = \mathbb{E}_{\hat{x}, \epsilon, t} \left[\|v_\phi(\hat{x}_t, t, \mathbf{z}_{:K}^{\text{txt}}) - (\epsilon - \hat{x})\|_2^2 \right]. \quad (8)$$

Training Objective. In representation pre-training, FLAT optimizes the total loss:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_{\text{txt}} \mathcal{L}_{\text{txt}} + \lambda_{\text{img}} \mathcal{L}_{\text{img}} \quad (9)$$

In fine-tuning, we show that each task-specific objective can be optimized separately over one of the encoder and decoders to maximize model performance.

Table 1 Text-to-image synthesis on GenEval. FLAT achieves competitive scores while enabling inference-time sequence length selection. Evaluated from a single fine-tuned checkpoint across prefix lengths. † marks a system that rewrites the prompt before generating.

Method	Backbone	GenEval
Chameleon (Chameleon Team, 2024)	scratch 7B	0.39
Janus (Wu et al., 2024a)	DeepSeek 1.5B	0.61
Transfusion (Zhou et al., 2025)	scratch 7B	0.63
JanusFlow (Ma et al., 2024)	DeepSeek 1.5B	0.63
Emu3† (Wang et al., 2024)	scratch 7B	0.66
Show-o-512 (Xie et al., 2024)	Phi-1.5 1.3B	0.68
Janus-Pro-1B (Chen et al., 2025b)	DeepSeek 1.5B	0.73
MetaQuery-L (Pan et al., 2025)	Qwen2.5-VL 3B	0.78
Janus-Pro-7B (Chen et al., 2025b)	DeepSeek 7B	0.80
MetaQuery-XL† (Pan et al., 2025)	Qwen2.5-VL 7B	0.80
FLAT, $K=1$	Qwen3.5 2B	0.49
FLAT, $K=4$	Qwen3.5 2B	0.77
FLAT, $K=16$	Qwen3.5 2B	0.83
FLAT, $K=64$	Qwen3.5 2B	0.83
FLAT, $K=256$	Qwen3.5 2B	0.82

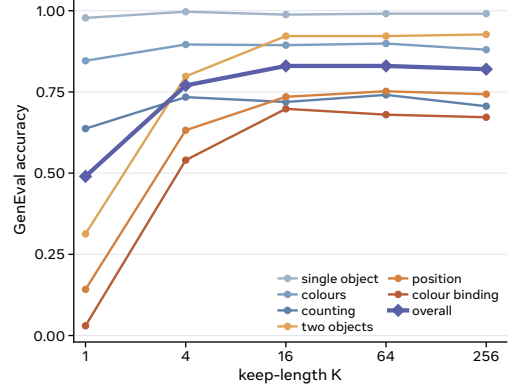


Figure 2 GenEval accuracy across keep-lengths broken down by eval category. Smaller K can capture core entity and color, but larger K is needed to capture spatial relations, multi-object layouts and attribute bindings.

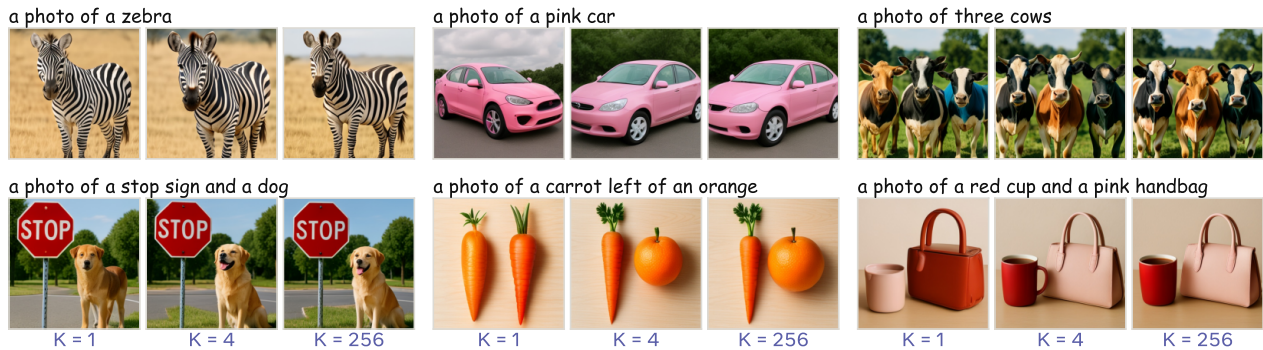


Figure 3 Qualitative T2I synthesis across different prefix lengths. Large K provides more fine-grained visual details.

4 Experiments

We evaluate FLAT across generation, retrieval, and probing tasks to assess its unified 1D continuous representations. Our evaluation spans three core dimensions: (1) **Generative Evaluation**, demonstrating that the FLAT representation drives both T2I synthesis and I2T captioning competitively against specialized baselines; (2) **Discriminative Evaluation**, evaluating the FLAT representation directly with cross-modal retrieval, linear probing, and unsupervised clustering; and (3) **Geometry Analysis**, analyzing latent geometry to show that FLAT projects image and text into a unified representation space, natively unlocking emergent zero-shot linear interpolation, latent space arithmetic and composed retrieval. App. G further extends this evaluation to zero-shot generalization across multilingual prompts and video modalities.

4.1 Implementation Details

The representation encoder and text decoder are built on a Qwen3.5-2B backbone (Qwen Team, 2026) with different LoRA adapters, while the image decoder is initialized from SANA-1.6B (Xie et al., 2025). Full architectural, pre-training, and task-specific adaptation configurations are provided in App. B, together with a summary of the fine-tuning recipes in Table 7. All evaluations are conducted as a function of prefix length K . Specifically, we benchmark T2I synthesis on GenEval (Ghosh et al., 2023) without prompt rewriting; I2T captioning on the MS-COCO Karpathy test split; cross-modal retrieval on both the MS-COCO and Flickr30k Karpathy splits (Karpathy and Fei-Fei, 2015); and composed retrieval on the MMEB CIRRR test split (Jiang et al., 2025; Liu et al., 2021) (see App. B.1 for full eval protocols). While our main result focuses on the final fine-tuned checkpoints, we report pre-trained, zero-shot performance across all tasks in App. C and task-only

Table 2 Image captioning on the MS-COCO Karpathy test split. FLAT matches or outperforms competitive baselines with less latent features. Column *Latent Feat.* represents latent feature shape which the text decoder is conditioned on. The metrics follow previous work: B@4 (BLEU-4), M (METEOR), R (ROUGE-L), C (CIDEr), S (SPICE).

Method	Latent Feat.	B@4	M	R	C	S
CrossFlow (Liu et al., 2025)	$4 \times 32 \times 32$	36.4	27.8	57.1	116.2	20.4
SCD-Net (Luo et al., 2023)	$N \times 512$	37.3	28.1	58.0	118.0	21.6
Florence-2-L (Xiao et al., 2024)	—	—	—	—	143.3	—
CoCa (Yu et al., 2022)	256×1408	40.9	33.9	—	143.6	24.7
BLIP-2 (Li et al., 2023b)	32×768	43.7	—	—	145.8	—
FLAT, $K=1$	1×64	35.3	28.7	57.7	122.0	22.0
FLAT, $K=4$	4×64	38.1	29.9	59.4	131.0	23.3
FLAT, $K=16$	16×64	39.6	30.8	60.4	136.2	24.1
FLAT, $K=64$	64×64	40.2	31.1	60.8	138.1	24.4
FLAT, $K=256$	256×64	40.5	31.2	60.8	138.6	24.4

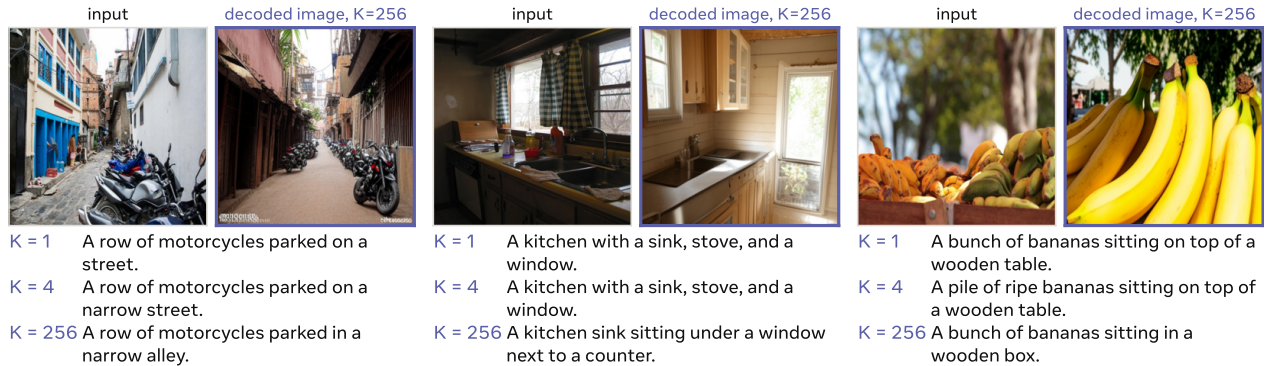


Figure 4 Image captioning with fine-tuned I2T decoder. Small K generates semantically meaningful captions and larger K refines lexical details.

adaptation without FLAT pre-training in App. D.

4.2 Generative Evaluation

4.2.1 Text-to-Image Generation

For T2I generation, we continue training the image decoder from the pre-trained checkpoint for 8k steps on 120K clean image-text pairs (Table 7). As shown in Table 1, FLAT achieves a GenEval score of **0.83** using a 2B encoder without prompt rewriting, compared with 0.78 for MetaQuery-L under the same no-rewrite protocol. Categorical GenEval analysis further demonstrates a clear link between task complexity and required prefix length K (Fig. 2). While a single token ($K=1$) captures basic semantics like single object and color, relational and compositional tasks require larger prefixes. Complex categories—two objects, position, and color binding—score near zero at $K=1$ (e.g., 0.03 for color binding), but recover steeply by $K=4$ and $K=16$ (reaching 0.67). As shown in Fig. 3, $K=1$ generates only the main entity, while spatial arrangements, secondary entities, and attribute-object bindings emerge accurately as K increases.

4.2.2 Image-to-Text Generation

The I2T evaluation fine-tunes only the I2T-decoder LoRA with captioning loss on COCO Karpathy training+restval set for 5k steps (Table 7). As detailed in Table 2, performance improves steadily across all metrics as K increases. Qualitatively (Fig. 4), a single token ($K=1$) generates well-formed, coherent captions, while additional tokens enhance lexical precision (e.g., street $\rightarrow$ alley) and concrete detail (e.g., a shelf $\rightarrow$ a shelf with baskets). These qualitative refinements directly mirror the quantitative gains in Table 2. Even at $K=1$, FLAT outperforms baselines such as CrossFlow (Liu et al., 2025) and SCD-Net (Luo et al., 2023) across

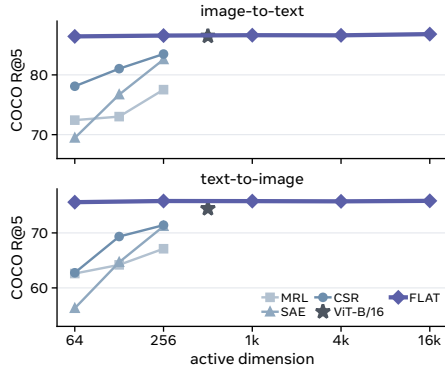


Figure 5 MS-COCO retrieval accuracy across active dimensions. Unlike other variable-width baselines whose performance degrades sharply at reduced dimensions, FLAT maintains stable retrieval accuracy across all dimensions.

Table 3 T2I and I2T retrieval results across prefix K . The first token captures key cross-modal discriminability, keeping retrieval accuracy virtually unaffected by prefix truncation. Performance is evaluated on the MS-COCO and Flickr30k Karpathy test splits.

K	I→T			T→I		
	R@1	R@5	R@10	R@1	R@5	R@10
<i>MS-COCO Karpathy 5k</i>						
1	63.00	86.44	92.50	47.76	75.59	84.67
4	63.40	86.58	92.58	47.93	75.80	84.71
16	63.30	86.64	92.56	47.94	75.78	84.72
64	63.32	86.62	92.56	47.73	75.74	84.69
256	63.54	86.82	92.62	47.98	75.83	84.78
<i>Flickr30k Karpathy 1k</i>						
1	90.50	98.10	99.50	76.94	93.52	96.48
4	91.20	98.30	99.50	77.40	93.58	96.54
16	90.80	98.30	99.60	77.32	93.60	96.54
64	90.90	98.20	99.40	76.88	93.72	96.54
256	90.80	98.20	99.30	76.26	93.16	96.40

CIDEr, METEOR, and SPICE scores.

4.3 Discriminative Evaluation

4.3.1 Cross-Modal Retrieval

For each benchmark, retrieval adapts the encoder LoRA, registers, and latent projection using contrastive loss on the corresponding training split. The reported checkpoints use 5k steps for COCO and 320 steps for Flickr30K (Table 7). Fig. 5 compares FLAT against representations that also support adaptive dimensionality from Wen et al. (2025), including MRL, SAE, and CSR. For FLAT, each token contains 64 dimensions, spanning from a 64-dimensional embedding ($K=1$) to a 16,384-dimensional embedding ($K=256$). Notably, FLAT’s retrieval performance remains virtually invariant across varying K , whereas baseline accuracy degrades sharply as K decreases. Table 3 reports detailed retrieval results across the K spectrum. Remarkably, a single token is sufficient for retrieval: every metric stays within roughly one percentage point across a $256\times$ reduction in width. This mirrors our observations in T2I and I2T generation, confirming that the first token carries sufficient global semantics and discriminative power.

4.3.2 Linear Probing and Clustering

To directly evaluate the representation space, we perform linear probing and unsupervised clustering on FLAT representations. A single linear classifier trained on a frozen $K=1$ token achieves 73.3% top-1 accuracy, scaling to 81.2% at $K=16$. This outperforms all purely generative latent spaces and surpasses DREAM (Li et al., 2026) which jointly trains the representation encoder with an image decoder, at equivalent dimensionality. Furthermore, without fitting any parametric heads, k -means clustering on a single FLAT token cleanly recovers clusters aligned with ImageNet classes (Fig. 6).

4.4 Geometry Analysis

4.4.1 Geometry Visualization

As shown in Fig. 7, while dual-encoder models like CLIP and SigLIP2 suffer from a persistent modality gap despite strong retrieval performance (Radford et al., 2021; Tschannen et al., 2025; Liang et al., 2022), FLAT projects both modalities into a tighter embedding space. FLAT achieves significantly greater cross-modal overlap: a single register token cuts CLIP’s centroid distance by half, and the full 256 tokens reduces it to a quarter. This tightly aligned geometry enables zero-shot latent operations as below.

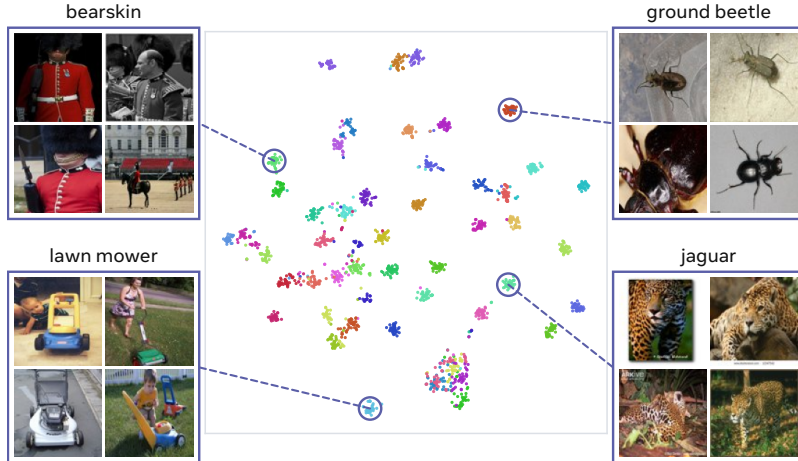


Figure 6 Unsupervised k -means clustering with a single FLAT token. Clusters formed from frozen $K=1$ representations (64 dimensions) align with ImageNet ground-truth classes without label supervision.

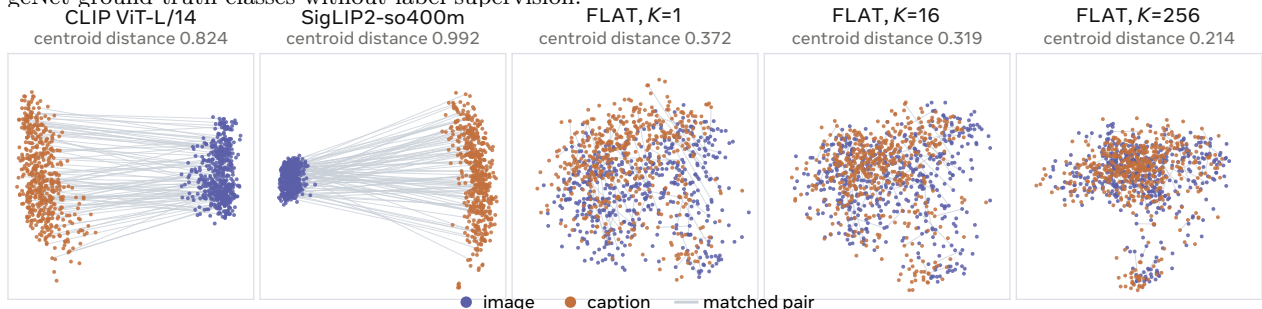


Figure 7 PCA distributions for image-text pairs from MS-COCO. Dual-encoder baselines exhibit clear spatial separation between image and text representations, whereas FLAT projects both modalities into a highly overlapping latent space. Increasing prefix K further shrinks the cross-modal centroid distance.

4.4.2 Continuous Feature Interpolation

We perform linear interpolation between a pair of FLAT codes as $z_\alpha = (1 - \alpha)z_1 + \alpha z_2$, and decode each intermediate latent representation using both generative heads. As shown in Fig. 8, this yields smooth semantic transitions across the input image-text pairs. Both the image and text decoders generate intermediate concepts along the interpolation trajectory. Furthermore, the prefix length K dictates transition granularity: longer prefixes smoothly vary fine-grained attributes, whereas $K=1$ forms a coarser semantic average (App. F.4).

4.4.3 Latent Space Arithmetic

FLAT representations naturally support zero-shot arithmetic, $z_{\text{edit}} = z_1 - z_2 + z_3$, without explicit editing supervision. As shown in Fig. 9, subtracting *sunrise* and adding *a full moon at night* modifies the focal concept while preserving the surrounding scene context, with both decoders interpreting the composite representation consistently. Furthermore, these arithmetic operations can directly serve as queries for zero-shot composed image retrieval on CIRRR (App. G.3).

4.5 Ablation Study on Training Loss Combinations

FLAT jointly optimizes a contrastive loss alongside bidirectional generation losses for I2T captioning and T2I synthesis. To evaluate the contribution of each objective, we train ablation variants across all seven non-empty loss combinations for 40k steps. Figure 10 and Table 5 present these variants along with their normalized and absolute eval scores at $K=256$. The full objective achieves strong performance across all

Table 4 ImageNet-1K linear probing top-1 accuracy. FLAT outperforms both generative latent spaces and joint generative-contrastive models on ImageNet-1K linear probing. We evaluate this performance by fitting a single linear classifier on top of frozen, concatenated K tokens.

Method	Dim	Top-1
<i>Generative Latent Spaces</i>		
SD-VAE latent	4	8.0
REPA (Leng et al., 2025)	–	62.5
MAE-B (He et al., 2022)	768	68.0
<i>Jointly Contrastive and Generative</i>		
DREAM (Li et al., 2026)	1024	72.7
FLAT, $K=1$	64	73.3
FLAT, $K=4$	256	77.0
FLAT, $K=16$	1024	81.2
FLAT, $K=64$	4096	81.8
FLAT, $K=256$	16384	81.8



Figure 8 Continuous cross-modal feature interpolations. Linear interpolation between a pair of image–text produces smooth semantic transitions. Image and text decoders map the same representation to corresponding concepts.

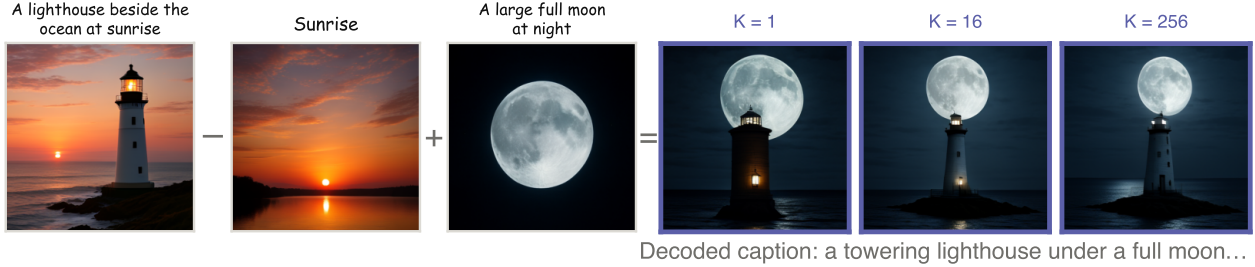


Figure 9 Zero-shot token-wise arithmetic operation. Linear addition and subtraction modify the target concept in both image and text outputs while preserving the surrounding scene.

five metrics, demonstrating that the symmetric contrastive and generative losses do not conflict, but rather mutually reinforce one another.

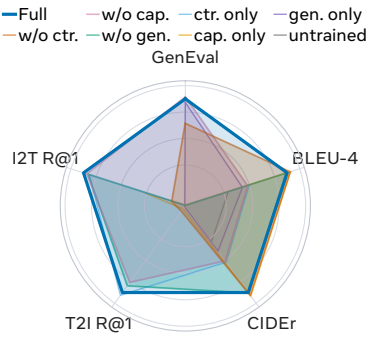


Figure 10 Eval for different training loss combinations (normalized scores).

Table 5 Eval for different training loss combinations (absolute scores).

Losses	Retrieval		Captioning		Generation
	I2T	T2I	B@4	CIDEr	GenEval
Full	52.2	56.3	40.2	137.5	0.327
w/o contrastive	13.6	6.7	40.7	139.2	0.297
w/o captioning	51.5	53.6	33.9	117.1	0.329
w/o generation	51.2	54.5	40.0	137.8	0.003
ctr. only	51.0	57.0	34.2	118.0	0.003
cap. only	39.4	4.0	40.8	139.7	0.004
gen. only	0.6	1.1	33.1	110.8	0.323
untrained	0.2	0.0	30.8	105.0	0.000

We further quantify how the three losses reinforce one another using a Shapley value decomposition (Shapley, 1953) over all eight coalitions. The attribution matrices and prefix sweeps in App. H show that while each task is driven primarily by its matched loss, the other objectives generally provide complementary signals.

5 Conclusion

We introduce FLAT, a representation pre-training framework that resamples images and text into 1D flexible-length, aligned, transmodal tokens. FLAT jointly optimizes a shared representation encoder alongside T2I and I2T decoders using bidirectional alignment and generation losses. This setup learns unified semantic representations not only useful for cross-modal retrieval but also as effective generative conditions for both decoders. Empirical results demonstrate that FLAT natively supports variable-length retrieval and synthesis, and with lightweight task-specific fine-tuning pushing T2I / I2T retrieval and generation performance to SOTA levels. Furthermore, qualitative analysis reveals that FLAT tokens enable zero-shot linear interpolation, latent space arithmetic, and composed retrieval.

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Appendix

A Overview

This appendix first details the implementation and evaluation protocols (Appendix B), followed by zero-shot results from the pre-trained checkpoint (Appendix C) and controlled task-only adaptation comparisons (Appendix D). We then provide expanded qualitative galleries (Appendix E), analyze flexible-length representations and sampling choices (Appendix F), examine zero-shot transfer and latent composition (Appendix G), and report loss and truncation ablations (Appendices H and I).

B Implementation and evaluation details

FLAT is jointly pre-trained on 65.7M image-text pairs for 135k steps on 8 nodes using the config in Table 6. Table 7 summarizes the task-specific fine-tuning configs for checkpoints evaluated in Sec. 4

B.1 Evaluation protocol

T2I Generation. We evaluate text-to-image (T2I) generation using GenEval (Ghosh et al., 2023) across all 553 prompts, generating 4 images per prompt with 20 sampling steps and a classifier-free guidance (CFG) scale of 4.5. To ensure controlled comparisons across prefix lengths, the initial noise vector is sampled once per prompt and reused. We do not apply prompt rewriting at any stage.

I2T Generation. We evaluate image captioning (I2T) on the MS-COCO Karpathy test split using greedy decoding. Generated captions are evaluated via `pycocoevalcap`, reporting standard metrics: BLEU-4 (B@4), METEOR (M), ROUGE-L (R), CIDEr (C), and SPICE (S).

- **BLEU-4 (B@4):** Measures n -gram precision (up to 4-grams) between generated and reference captions, incorporating a brevity penalty to discourage overly short outputs.
- **METEOR (M):** Computes an F -score based on unigram precision and recall, accounting for word stemming, synonymy, and paraphrase alignment.
- **ROUGE-L (R):** Evaluates structural word-order similarity using the Longest Common Subsequence (LCS) between predicted captions and ground-truth references.
- **CIDEr (C):** Computes TF-IDF-weighted n -gram cosine similarity to reward visually descriptive content while penalizing generic terms.
- **SPICE (S):** Converts captions into semantic scene graphs (objects, attributes, and relations) and calculates the F_1 -score over the extracted graph tuples.

Retrieval. We evaluate both text-to-image (T2I) and image-to-text (I2T) retrieval on the COCO Karpathy 5K and Flickr30K Karpathy 1K test splits, reporting direct retrieval recall without post-hoc reranking.

Composed Retrieval. Following the MMEB protocol (Jiang et al., 2025), we evaluate on CIRRR (Liu et al., 2021) using 1,000 queries against a 1,000-image gallery. We perform latent space arithmetic by converting image editing instructions into ADD and REMOVE clauses. Specifically, let $\mathbf{z}_{\text{ref}}$ denote the original image representation and $\mathbf{z}_0$ denote the representation of the neutral phrase "a photo". For an edit specified by added phrases A and removed phrases R , the composite query representation is formulated as:

$$\mathbf{z}_q = \mathbf{z}_{\text{ref}} + \alpha \left[\sum_{a \in A} (\mathbf{z}_a - \mathbf{z}_0) - \sum_{r \in R} (\mathbf{z}_r - \mathbf{z}_0) \right]$$

After composition, we ℓ_2 -normalize each token representation and score the similarity with the late-interaction score defined in Eq. 3. We sweep $\alpha \in \{0.25, 0.5, 1, 1.5, 2, 3, 4, 6, 8\}$ for each K on this diagnostic set. Table 14 reports all metrics using the α selected by Recall@1.

Table 6 FLAT model and pre-training configuration.

Component	Configuration
Encoder and text decoder	
Model	Qwen3.5-2B, bfloat16, FlashAttention-2, gradient checkpointing
LoRA	$r=16$, $\alpha=32$, dropout 0.05; on Q/K/V/O and gate/up/down projections
Adapters	Two adapters sharing this configuration: encoder and caption decoder
Base weights	Frozen; 11.4M trainable across the adapter and register projections
Register sequence	
Register tokens	$N = 256$, latent dimension $d = 64$
Nested dropout	$p = 1.0$, $\mathcal{K} = \{1, 4, 16, 64, 256\}$ uniform; one shared K per optimization step
Truncation	Tail dropped, not null-padded, so a prefix is a shorter sequence (App. I)
Minimum tokens	1
Read-out head	MLP with norm matching and latent normalization
Quantizer	None; the representation stays continuous and is scored directly
Objectives	
Loss weights	Contrastive 1.0; text generation 1.0; image generation 1.0
Contrastive objective	InfoNCE, temperature 0.07, cross-device negatives
Image conditioning	Text registers at every step (<code>imagegen_image_cond_prob=0</code>)
Encoder instruction	<i>Represent the user’s input.</i>
Caption instruction	<i>Describe the input.</i>
Image decoder	
Model	SANA-1600M at 512 px, fully fine-tuned
Connector	16 Transformer layers, caption channels 2304, input scale 2.3452
Flow matching	Flow objective, shift 3.0, logit-normal t with mean 0 and std 1
CFG dropout	0.1
Data	
Image resolution	512×512 , giving a 16×16 grid of 256 visual tokens
Maximum text length	128 tokens
Training data	67M image–text pairs
Approx. composition	Long-caption (30M), short-caption (33M), prompt-style (4M), and curated high-quality (0.1M) pairs
Data workers	20 per rank, prefetch factor 6
Optimization	
Optimizer	AdamW, $\beta = (0.9, 0.999)$, weight decay 0.0, gradient clip 1.0
Learning rate	1.0×10^{-4} ; 500-step linear warm-up then cosine decay to zero
Batch size	16 per rank $\times$ 64 ranks $\times$ 1 accumulation step = 1024 global
Hardware	8 nodes $\times$ 8 H200 GPUs
Precision	bfloat16
Training steps	135k pretraining

Table 7 Task-specific fine-tuning configs. Each column describes the config of the task-specific fine-tuned checkpoint evaluated in Sec. 4. All runs use AdamW, bfloat16, zero weight decay, direct prefix truncation, and uniform sampling over $\mathcal{K} = \{1, 4, 16, 64, 256\}$. Batch sizes are global; retrieval uses the global batch as cross-device in-batch negatives.

Setting	T2I synthesis	I2T captioning	Image-text retrieval	
			COCO	Flickr30k
Training data	120K high-quality image-text pairs	COCO Karpathy train+restval (113,287 images)	COCO Karpathy train+restval	Flickr30k Karpathy train
Updated parameters	Image decoder	Text decoder LoRA	Encoder LoRA, registers, and latent projection	Encoder LoRA, registers, and latent projection
Objective	T2I generation	I2T generation	T2I an I2T contrastive	T2I an I2T contrastive
Checkpoint step	8,000	5,000	5,000	320
Global batch	1,024	256	2,048	2,048
Learning rate	4×10^{-4}	2×10^{-4}	1×10^{-4}	1×10^{-4}
Warm-up steps	5,000	2,213	500	128

Linear Probing. For linear probing, we train a linear classification head on top of the frozen, concatenated prefix- K token representations. We train on all 1,281,167 ImageNet-1K training images for 20 epochs and evaluate performance on the 50,000-image validation split.

C Evaluation Results of the pre-trained checkpoint

The jointly pre-trained checkpoint supports generation and retrieval without task-specific adaptation. We first evaluate generation across both modalities, followed by bidirectional retrieval.

T2I Generation. Table 8 reports GenEval results using the same prompt, sampling, and fixed-noise protocol as the fine-tuned checkpoint. Without task-specific tuning, the pre-trained model reaches an overall score of 0.71, with performance scaling from 0.32 using a single token to 0.71 with 256 tokens.

Table 8 GenEval for the pre-trained checkpoint. No prompt rewriting is applied.

K	1	4	16	64	256
GenEval	0.317	0.613	0.703	0.701	0.711

I2T Generation. We evaluate the pre-trained checkpoint directly on I2T captioning using the MS-COCO Karpathy test split. As shown in Table 9, performance steadily improves with larger prefix lengths, reaching 63.8 CIDEr and 16.3 SPICE at $K=256$. Although these absolute scores are lower than those of the fine-tuned checkpoint (Table 2), this gap is largely driven by stylistic differences of training data: the pre-training corpus contains diverse descriptive text, whereas COCO favors short, literal captions. This distributional gap inherently penalizes overlap-based metrics, as semantically correct but longer or editorial descriptions introduce phrasing that diverges from compact reference captions.

Figure 11 qualitatively illustrates this phenomenon. Zero-shot generations reliably capture the central entities and actions, but often express them as detailed descriptions or title-style captions. For instance, at $K=64$, 99.6% of generated sequences properly terminate with an EOS token and average 10.8 words, yet 27.1% begin with quotation marks characteristic of image titles. These results indicate that while pre-trained representations already encode rich visual semantics, fine-tuning primarily serves to align decoder length and stylistic tone with COCO expectations. Consequently, part of the metric gains observed after fine-tuning reflects output-format alignment rather than improved semantic understanding alone.



COCO reference

A lone man sitting on a bench on a hill next to a large tree.

Zero-shot, $K = 64$

"Solitary figure under the tree, seeking solace in nature."

Fine-tuned, $K = 64$

A person sitting on a bench under a tree.



COCO reference

A painting of a table with fruit on top of it.

Zero-shot, $K = 64$

"Still life with fruit, pitcher, and candle in ornate frame."

Fine-tuned, $K = 64$

A painting of a pitcher, oranges and a candle.



COCO reference

A commercial airplane ascending into the sky.

Zero-shot, $K = 64$

A sleek airplane soars through a clear sky, leaving a trail of light.

Fine-tuned, $K = 64$

A large jetliner flying through a gray sky.



COCO reference

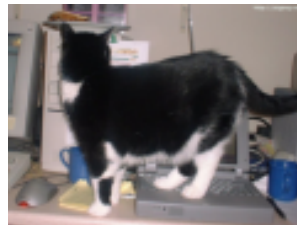
A black and white photo of a train sitting in a station.

Zero-shot, $K = 64$

"Classic train at station, black and white."

Fine-tuned, $K = 64$

A black and white photo of a train at a station.



COCO reference

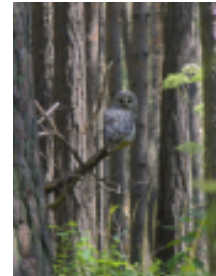
A black and white cat standing on a laptop.

Zero-shot, $K = 64$

A black and white cat stands on a laptop keyboard in an office setting.

Fine-tuned, $K = 64$

A black and white cat standing on a laptop.



COCO reference

An owl sitting on a branch in a forest.

Zero-shot, $K = 64$

A curious owl perches on a tree branch in a forest setting.

Fine-tuned, $K = 64$

An owl sitting on a branch in a forest.

Figure 11 A comparison between the generated captions between pre-trained and fine-tuned checkpoints. Each example compares a COCO reference caption against outputs from the pre-trained and fine-tuned checkpoints. While zero-shot predictions from the pre-trained model reliably capture core entities and actions, fine-tuning aligns output length and phrasing with the COCO annotation style.

Table 9 Zero-shot I2T captioning evaluation for the pre-trained checkpoint. Greedy decoding is used to generate the captions.

K	B@4	M	R	C	S
1	11.0	17.9	37.2	46.1	12.5
4	13.4	20.2	39.8	57.2	14.6
16	14.5	21.3	41.3	62.1	15.7
64	14.9	21.6	41.6	63.5	16.2
256	15.0	21.7	41.7	63.8	16.3

Retrieval. Table 10 evaluates the pre-trained checkpoint on the full MS-COCO and Flickr30K Karpathy test splits without retrieval-specific fine-tuning. Even in this zero-shot setting, the learned representations effectively support bidirectional retrieval. However, zero-shot performance degrades slightly as the prefix length K increases. Task-specific fine-tuning substantially mitigates this sensitivity to K (Table 3).

Table 10 Zero-shot retrieval for the pre-trained checkpoint. Results are recall (%) on the COCO Karpathy 5k and Flickr30k Karpathy 1k test splits.

K	Image $\rightarrow$ text			Text $\rightarrow$ image		
	R@1	R@5	R@10	R@1	R@5	R@10
<i>MS-COCO Karpathy 5k</i>						
1	44.90	69.10	78.54	39.70	64.59	74.51
4	44.48	68.26	78.08	37.85	62.31	72.08
16	41.94	66.96	76.32	36.32	61.02	70.84
64	39.96	65.02	74.64	34.33	59.07	69.37
256	35.48	60.04	70.64	31.45	55.87	66.57
<i>Flickr30k Karpathy 1k</i>						
1	64.10	89.70	94.90	64.40	86.04	91.16
4	63.50	89.10	94.60	61.52	84.62	89.76
16	61.00	89.00	94.60	60.44	83.98	89.30
64	59.30	88.10	93.50	58.68	83.00	88.64
256	55.70	84.30	91.90	55.74	81.30	87.20

D Effect of FLAT pre-training on task adaptation

We compare task-specific adaptation with and without FLAT pre-training. The task-only controls use the same architecture, training data, optimization budget, and evaluation protocol as the corresponding fine-tuned FLAT models. They retain the public backbone initialization but begin with randomly initialized FLAT-specific parameters and receive no joint FLAT pre-training. Table 11 reports the complete prefix-length sweep. For T2I synthesis, the task-only model remains near 0.60 GenEval across all prefix lengths. FLAT pre-training improves GenEval by 0.16–0.24 for $K \geq 4$ and reaches 0.83 at $K=16$ and $K=64$. Although the task-only model is higher at $K=1$, its performance does not increase with additional tokens. In contrast, the pre-trained initialization gains 0.34 from $K=1$ to $K=16$, showing that pre-training enables the image decoder to exploit wider prefixes for compositional generation. Pre-training also improves captioning by 9.0–10.9 BLEU-4 and 31.9–41.2 CIDEr across K . Its effect on retrieval is smaller and direction-dependent: image-to-text R@1 improves by 0.6–1.3 points, while the task-only model is 0.4–0.9 points higher for text-to-image R@1. These results show that the generative tasks benefit most from the representation learned during joint pre-training, whereas in-domain contrastive adaptation learns most of the retrieval interface directly.

Table 11 Task-specific adaptation with and without FLAT pre-training. T2I uses matched 8k-step recipes on 120K image-text pairs; captioning and retrieval use matched 5k-step recipes on the COCO Karpathy train+restval split. T2I is evaluated on GenEval without prompt rewriting. Captioning and retrieval are evaluated on the Karpathy 5k test split, with full galleries for retrieval.

Metric	Initialization	$K=1$	$K=4$	$K=16$	$K=64$	$K=256$
T2I GenEval	Task-only	0.60	0.61	0.61	0.60	0.59
	FLAT pre-trained	0.49	0.77	0.83	0.83	0.82
Captioning B@4	Task-only	24.4	27.3	30.6	31.0	30.8
	FLAT pre-trained	35.3	38.1	39.6	40.2	40.5
Captioning CIDEr	Task-only	80.8	91.5	104.3	105.3	105.0
	FLAT pre-trained	122.0	131.0	136.2	138.1	138.6
I2T retrieval R@1	Task-only	62.44	62.38	62.28	62.72	62.24
	FLAT pre-trained	63.00	63.40	63.30	63.32	63.54
T2I retrieval R@1	Task-only	48.62	48.54	48.49	48.53	48.43
	FLAT pre-trained	47.76	47.93	47.94	47.73	47.98

E Qualitative results for FLAT

The following galleries present a qualitative evaluation of FLAT. Text-to-image samples share the same initial noise across prefix lengths, while retrieval evaluations utilize task-specific checkpoints and full galleries as described in Table 7. Figures 12 and 13 evaluate performance on controlled compositional prompts, whereas Figure 14 extends this comparison to longer, free-form prompts. Figure 15 illustrates bidirectional retrieval results, and Figure 16 highlights representative failure cases across the primary tasks.

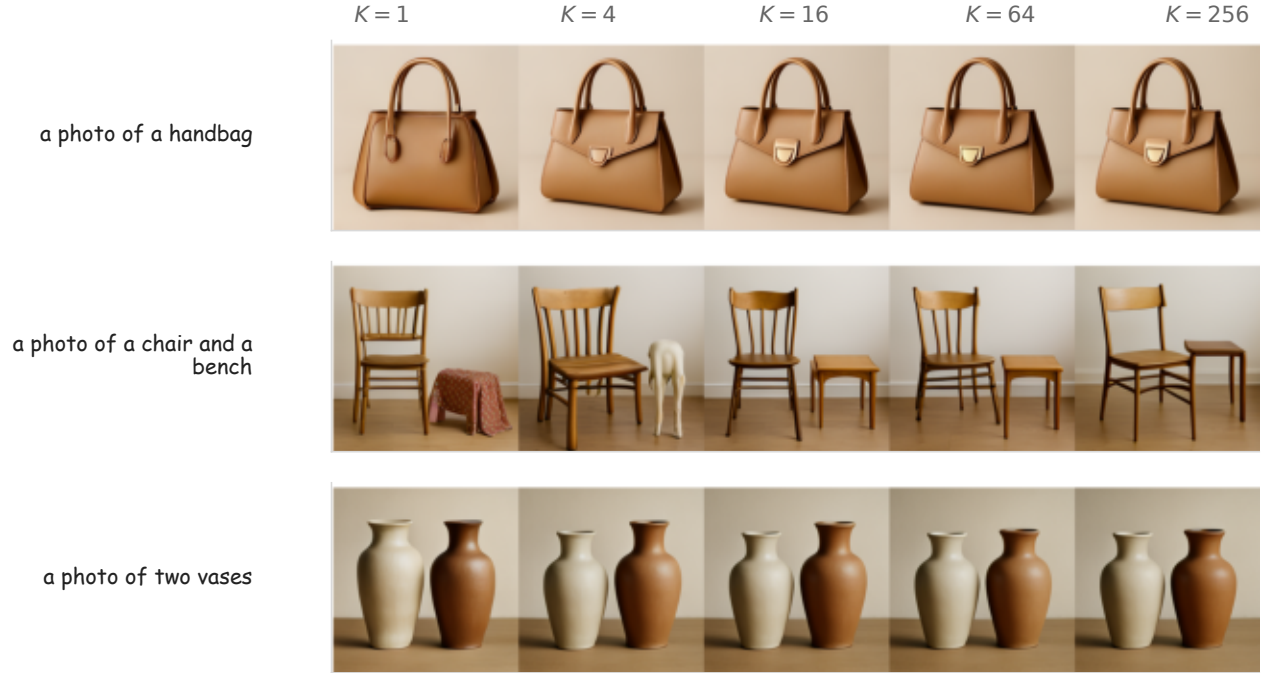


Figure 12 Text-to-image gallery, part I. Samples are drawn from the fine-tuned checkpoint across the full prefix-length ladder. The initial noise is fixed within each row. These examples cover a single object, two-object composition, and counting.

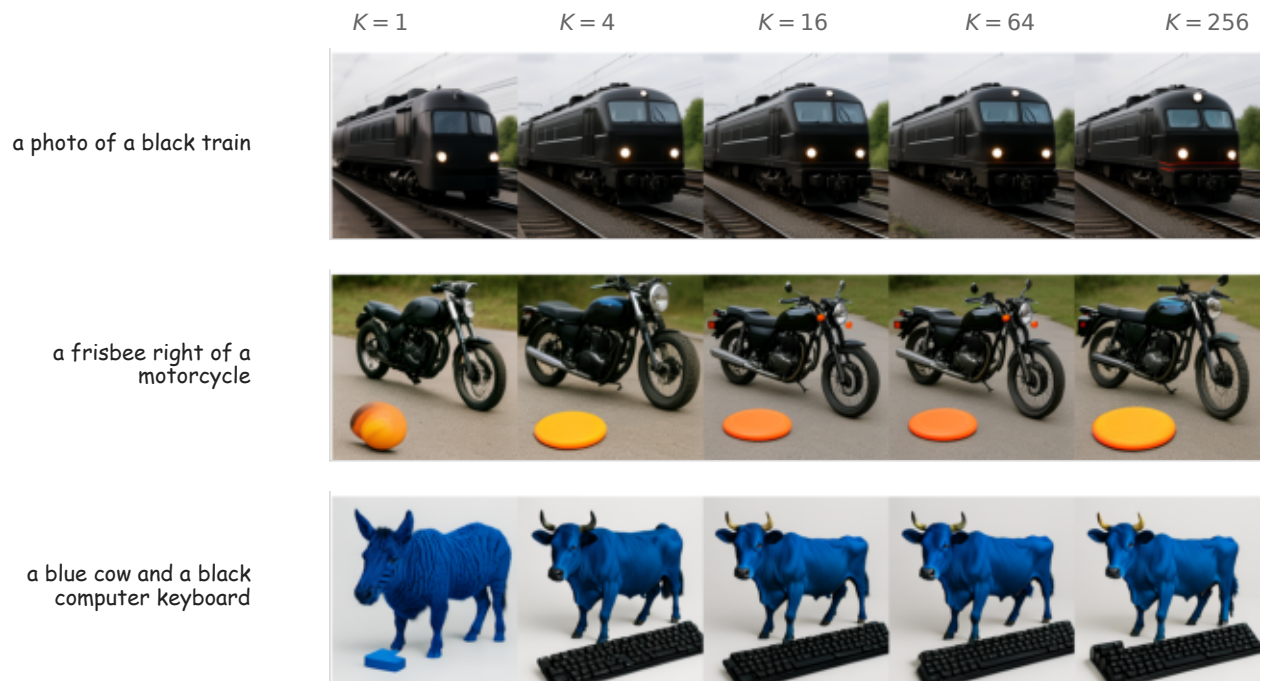


Figure 13 Text-to-image gallery, part II. Additional samples covering color, spatial relations, and attribute-object binding, with fixed initial noise across prefix lengths.

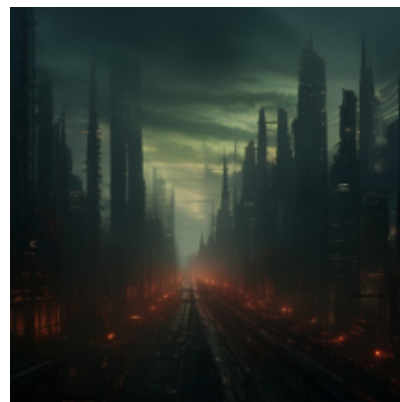
Fresh fruits, vegetables, and berries illuminated by golden morning sunlight, photographed with a Sony a7 III and an 85mm f/1.2 lens



A rocky seashore with sandy patches, yellow snail shells, seaweed, and pine pollen on the waves; a concrete war fortress with orange lichen in the distance at sunrise



A cinematic and ultrarealistic rendering of a futuristic city street at dawn, complete with atmospheric lighting and realistic textures



The father of Jesus Christ as a human being



Mexican Buffy the Vampire Slayer and the ancestral line of slayers, ink illustration, colorful, magical realism, HD



Orobanche ramosa on a black background, shot on Fuji film, surreal, award-winning photograph, 8K, rich details

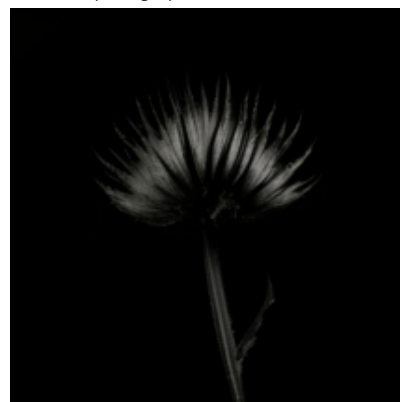


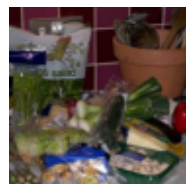
Figure 14 Text-to-image gallery with free-form prompts. Samples at $K=64$ for longer prompts from MJHQ-30K, spanning photographic and illustrative styles across diverse subjects and scenes.

(a) Image-to-text retrieval



query image
target: rank 1

- 1 A man with a red helmet on a small moped on a dirt road.
- 2 A person riding a motorcycle on a road with a hill in the background.
- 3 A dump truck passing a gentleman on a bicycle loaded with wood benches.



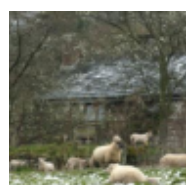
query image
target: rank 1

- 1 A bunch of groceries are piled onto a table.
- 2 A table topped with lots of vegetables of different color and kind.
- 3 A counter with vegetables, knife and cutting board on it.



query image
target: rank 3

- 1 A plane on the runway is being led by a tow cart.
- 2 A large jetliner sitting on top of an airport tarmac.
- 3 A large jetliner sitting on top of an airport runway.

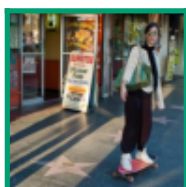


query image
target: rank 2

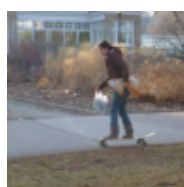
- 1 A sheep in the snow with other sheep around him.
- 2 A herd of sheep standing on top of snow covered field.
- 3 A bunch of lambs standing around in a field behind a fence.

(b) Text-to-image retrieval

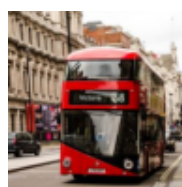
A girl is skateboarding down the Hollywood walk of fame.



rank 1 target



rank 2

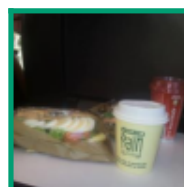


rank 3

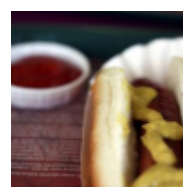
A sandwich wrapped in a bag, a to-go coffee cup and a drink sits on top of a white table.



rank 1



rank 2 target

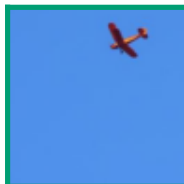


rank 3

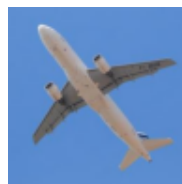
A small airplane flying through a blue sky.



rank 1

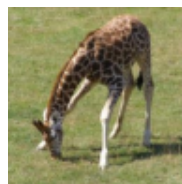


rank 2 target

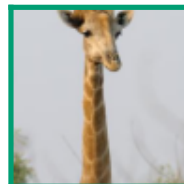


rank 3

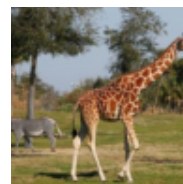
A giraffe standing in front of green brush.



rank 1



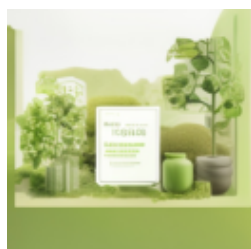
rank 2 target



rank 3

Figure 15 Bidirectional retrieval galleries. Image-to-text (top) and text-to-image (bottom) retrieval results at $K=16$ on MS-COCO. A green box highlights the ground-truth target, while remaining images/texts show top-ranked retrieved candidates. Cases where the ground truth appears at rank 2 or 3 illustrate near-miss failures, where leading alternatives remain semantically consistent with the query.

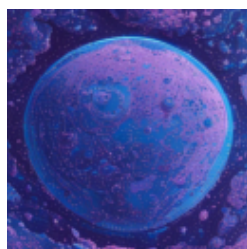
(a) Text-to-image synthesis



FLAT, $K = 64$

modern ecommerce landing page, agriculture, plants, organic, science, simple, clean, minimalist, 4k

Minor miss: rendered text remains indistinct.

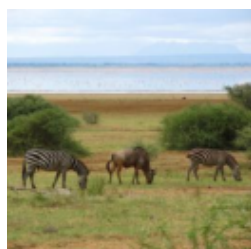


FLAT, $K = 64$

a satellite view of a planet pink blue and purple with lot of city on the surface, retrofuturism, in space, cartoon style

Minor miss: the city structure is only weakly expressed.

(b) Image captioning



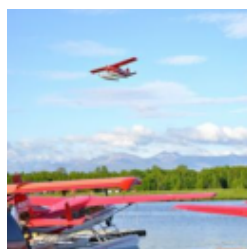
reference

An antelope is eating grass in between two zebra.

FLAT, $K = 64$

A herd of zebra standing on top of a grass covered field.

Minor miss: the antelope is omitted.



reference

A plane with water skis for landing gear coming in for a landing at a lake.

FLAT, $K = 64$

A small plane flying over a body of water.

Minor miss: the pontoons and landing action are omitted.

(c) Image-to-text retrieval



1 A plane on the runway is being led by a tow cart.

2 A large jetliner sitting on top of an airport tarmac.

3 A large jetliner sitting on top of an airport runway.

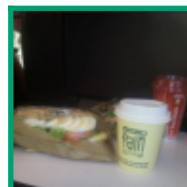
target: rank 3

(d) Text-to-image retrieval

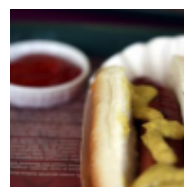
A sandwich wrapped in a bag, a to-go coffee cup and a drink sits on top of a white table.



rank 1



rank 2 target



rank 3

Figure 16 Representative failure cases across tasks. For text-to-image synthesis (top), the overall subject matter and style are preserved, though fine-grained layout or typography exhibits minor imperfections. For captioning (middle), the primary scene is described accurately, but secondary objects or actions are occasionally omitted. In bidirectional retrieval, ground-truth targets consistently rank near the top, preceded only by semantically close alternatives.

F Flexible-length representation analysis

During training, FLAT samples K only from a list $\mathcal{K} = \{1, 4, 16, 64, 256\}$. We examine whether the other choice of K can be flexibly applied at test time.

F.1 Generalization to unseen prefix lengths

In short, direct prefix truncation enables inference at arbitrary integer sequence lengths within 256. Figure 17 illustrates outputs generated from the same prompt and initial noise across both trained lengths and unseen intermediate lengths. Generations at unseen prefix lengths remain coherent and evolve smoothly with K .

A red passenger train crossing a stone bridge through snowy mountains at sunset



A silver robot cooking tomato soup in a modern kitchen, holding a wooden spoon



A traditional blue-and-white porcelain teapot beside two matching teacups on a bamboo tray



A red cube to the left of a blue sphere, with a small yellow cone centered above them, on a white background



A glass terrarium containing a miniature glowing forest and a tiny waterfall on a dark wooden table



Figure 17 Generation at trained and unseen prefix lengths. Blue borders indicate sequence lengths sampled during training, whereas gray borders denote unseen intermediate lengths. All columns share the same prompt, sampling configuration, and initial noise vector.

F.2 Prefix-dependent compute

Because the representation encoder is causal, the hidden states of the first K registers are independent of subsequent register positions. At inference time, we can therefore process only the first K registers rather than encoding all $N = 256$ registers and truncating post-hoc. Table 12 reports single-batch inference FLOPs for the encoder and both decoders, counting each multiply-add operation as two FLOPs. Reducing K from 256 to 1 reduces encoder and I2T decoder compute by 50.0% and 71.1%, respectively. In contrast, T2I

compute decreases by 13.8%, as its cost is overwhelmingly dominated by spatial denoising and VAE decoding rather than conditioning length.

Table 12 Encoder and decoder compute under prefix truncation. Absolute cost is reported in TFLOPs; relative cost normalizes each component to its $K=256$ setting.

K	Encoder		I2T decoder		T2I decoder	
	TFLOPs	Relative	TFLOPs	Relative	TFLOPs	Relative
1	0.714	50.0%	0.289	28.9%	29.964	86.2%
4	0.723	50.6%	0.298	29.7%	30.020	86.4%
16	0.756	52.9%	0.331	33.1%	30.245	87.1%
64	0.891	62.3%	0.465	46.4%	31.144	89.6%
256	1.430	100.0%	1.002	100.0%	34.741	100.0%

F.3 Prefix-length sampling strategies

We sample the prefix K from the ladder $\mathcal{K} = \{1, 4, 16, 64, 256\}$ using three strategies: 1) **Uniform** assigns equal probability to every K ; 2) **Geometric** samples K_r with $p(K_r) \propto \beta^r$, where $\beta = 1.3$; and 3) **Replay** spends half of training on shorter prefixes and the other half on the full prefix.

Figure 18 compares the training loss through 40k steps. Uniform and Geometric sample every K throughout training, whereas Replay progressively unlocks longer prefixes.

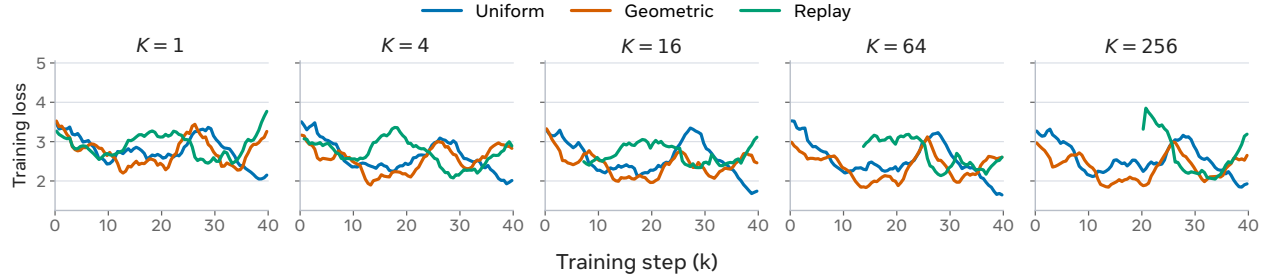


Figure 18 Training loss under different prefix-length sampling strategies. Training lasts 40k steps.

Metric	Sampling	$K=1$	$K=4$	$K=16$	$K=64$	$K=256$
GenEval ($\times 100$) $\uparrow$	Uniform	33.0	32.5	32.4	31.9	32.7
	Geometric	26.0	43.2	49.9	49.9	50.2
	Replay	32.2	32.6	32.2	32.0	32.3
I2T R@1 $\uparrow$	Uniform	57.8	53.9	54.3	52.9	52.2
	Geometric	56.1	52.6	53.1	52.2	51.8
	Replay	59.1	55.7	55.8	54.5	52.1
T2I R@1 $\uparrow$	Uniform	62.4	60.1	58.4	56.7	56.3
	Geometric	60.9	56.9	56.6	55.8	55.5
	Replay	62.2	60.1	59.7	59.2	59.1
CIDEr $\uparrow$	Uniform	122.7	130.4	136.8	137.6	137.5
	Geometric	123.0	129.6	137.0	137.3	138.3
	Replay	121.5	131.3	136.5	137.5	137.5

Table 13 Downstream performance under different prefix-length sampling strategies.

Table 13 evaluates matched 40k-step checkpoints. Geometric favors generation for $K \geq 4$, Replay favors retrieval. However, no single strategy is universally optimal across all metrics and values of K .

F.4 Latent interpolation

Figures 19 and 20 examine latent interpolation through both decoders. Across five prefix lengths, Figure 19 shows that longer prefixes transition multiple attributes smoothly, while shorter prefixes yield coarser blends. Figure 20 adds six $K=256$ walks spanning identity, appearance, structure, context, and season; captions are condensed to keywords.

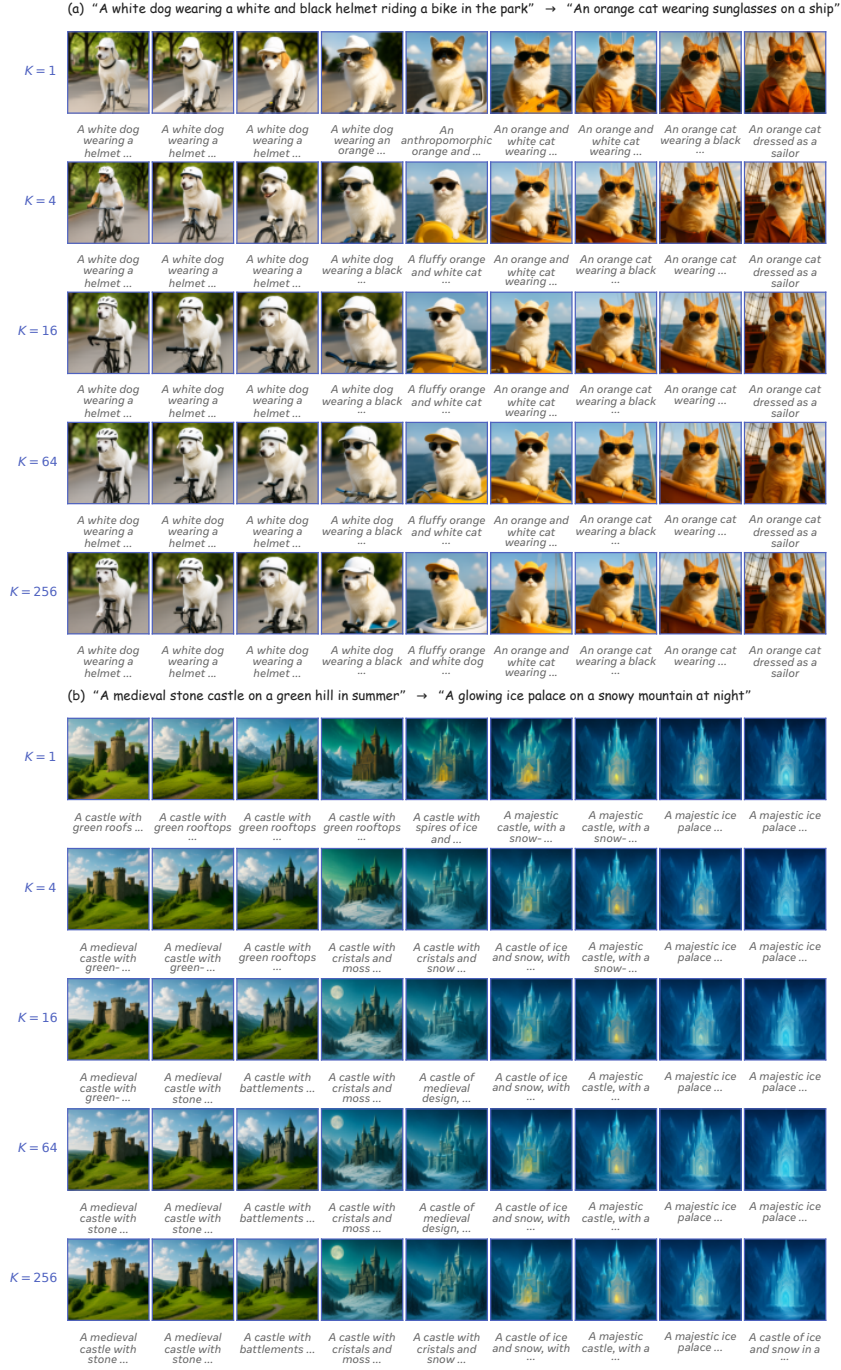


Figure 19 Latent walks across prefix lengths. Two image-text pairs are interpolated at $K \in \{1, 4, 16, 64, 256\}$. Italic text under every image is produced by the I2T decoder.

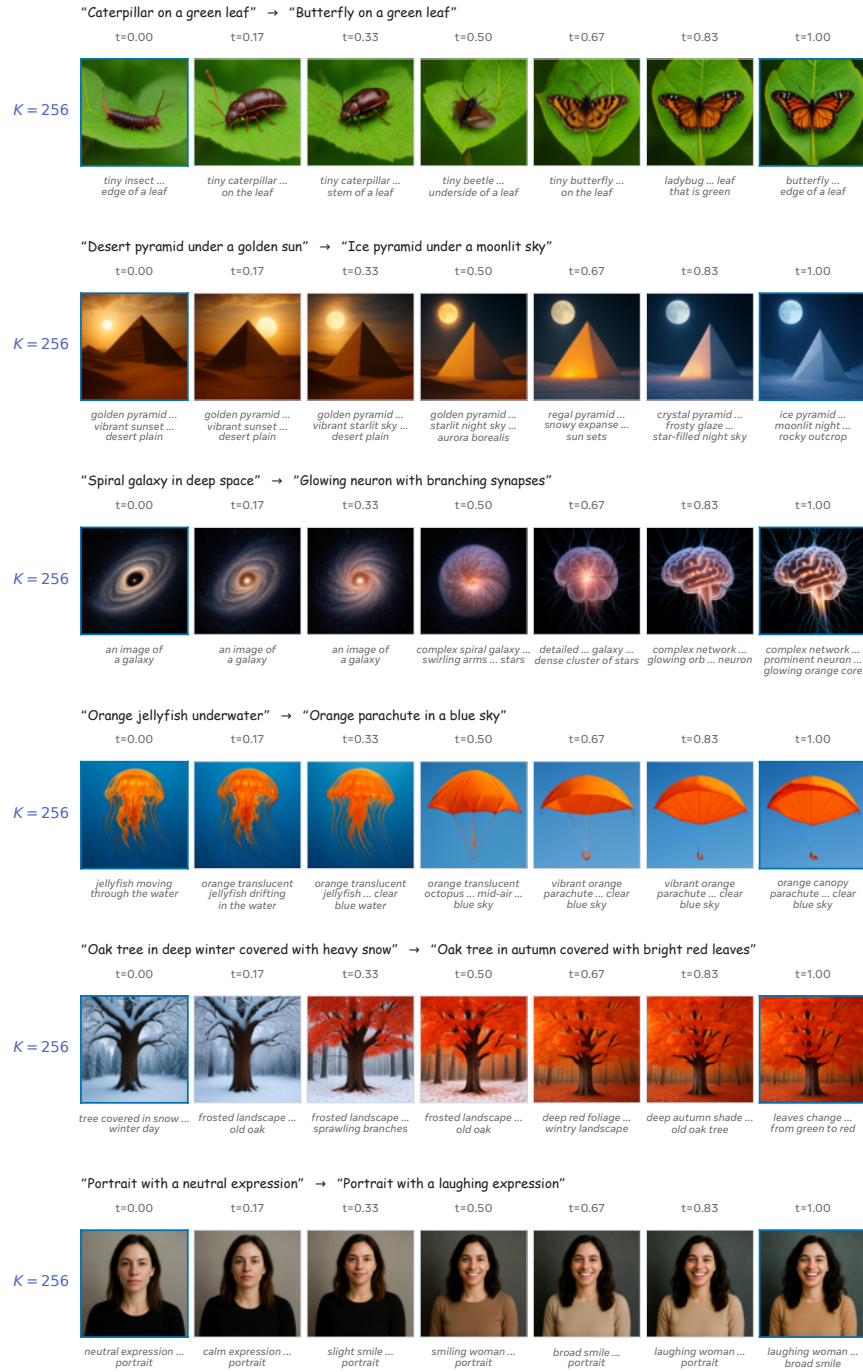


Figure 20 Additional latent walks at $K=256$. Blue borders mark the two endpoints. Italic text under every image is produced by the I2T decoder.

G Zero-shot transfer and latent composition

In this section, we demonstrate that the pre-trained FLAT representation natively supports zero-shot multilingual text understanding, multi-frame video understanding, and composed retrieval.

G.1 Multilingual text and emoji understanding

We first show that the representation encoder inherits multilingual capabilities from its LLM backbone. Consequently, non-English prompts occupy a similar representation space to English, despite the model being trained exclusively on English data. Figure 21 demonstrates that French and Chinese prompts decode into images consistent with their English counterparts, while emoji sequences reliably decode into their corresponding visual referents.

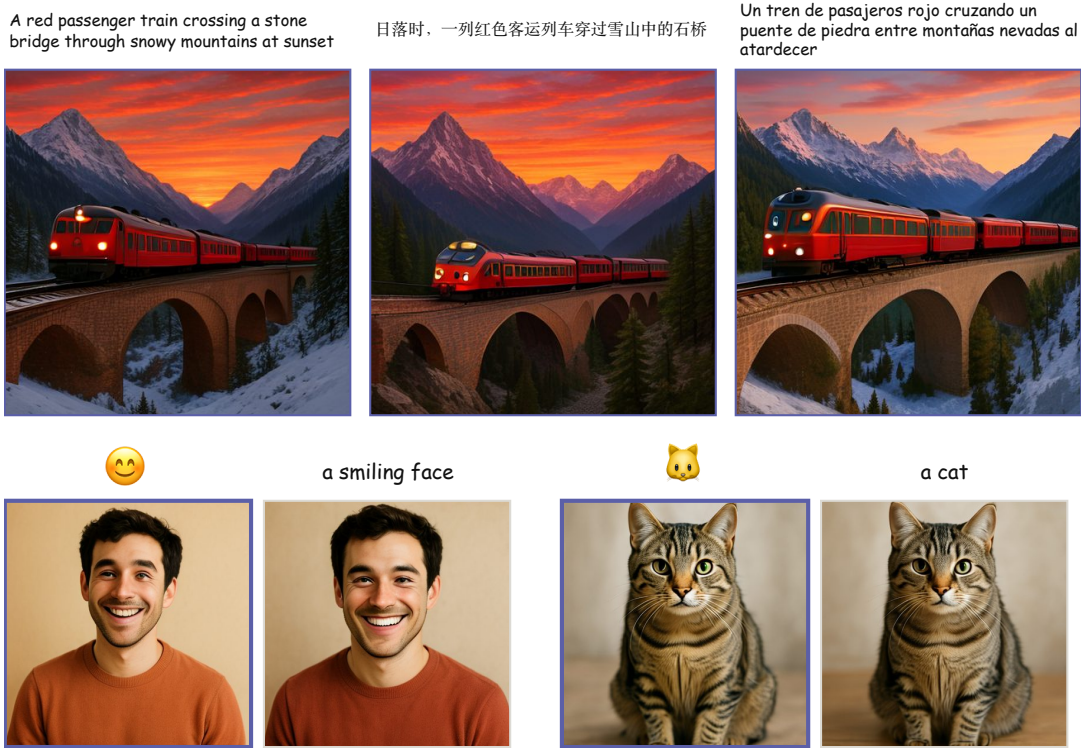


Figure 21 Zero-shot multilingual text and emoji conditioned image generation. Although trained exclusively on English data, FLAT generalizes its understanding to non-English languages and emojis.

G.2 Multi-frame video understanding

We next evaluate FLAT’s ability to process multi-frame video inputs despite being trained exclusively on static images. Specifically, we jointly encode 4 sampled frames using the representation encoder and decode the resulting sequence via the image generation head. Figure 22 shows examples from an aquarium scene and a cooking sequence. At $K=64$, the generated outputs preserve persistent objects and scene context while integrating information across the sampled frames. These results demonstrate that the FLAT representation naturally aggregates multi-frame context into a unified, thumbnail-like visual output.

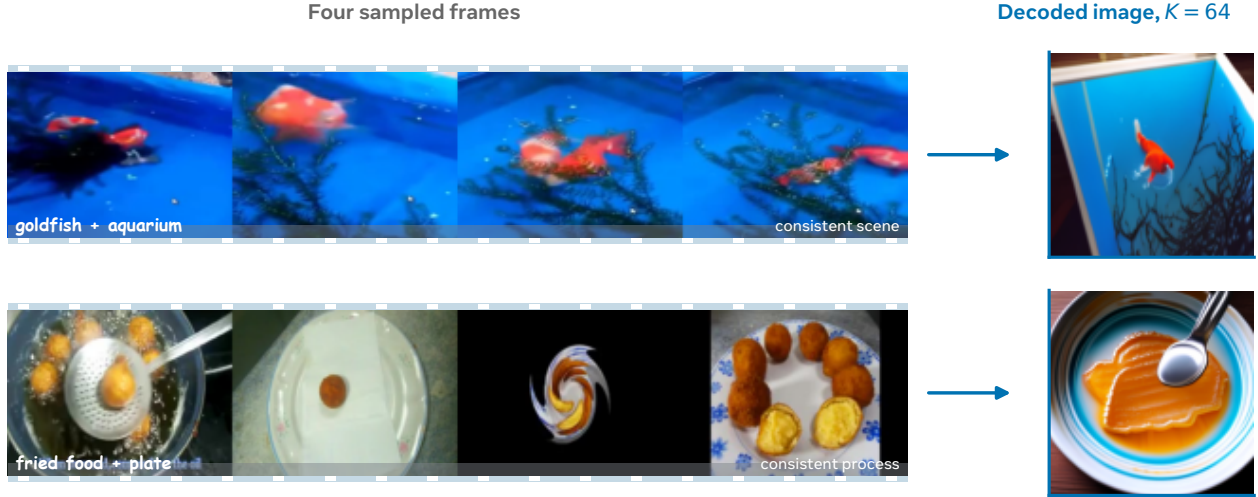


Figure 22 Zero-shot multi-frame video semantic aggregation. Each row jointly encodes 4 sampled frames and decodes one image at $K=64$. The generated outputs preserve persistent objects and scene context across the sampled frames.

G.3 Composed retrieval

We evaluate latent space arithmetic on the CIRRR dataset without training on composed queries, following the protocol outlined in Appendix B.1. As shown in Table 14, using a prefix length of $K=1$ yields the strongest performance, demonstrating that an edit vector applied to a single token representation can effectively steer the global representation space. Figure 23 shows the images retrieved with the raw image and editing instructions with FLAT zero-shot.

Table 14 Zero-shot CIRRR composed-retrieval diagnostic across keep-lengths. Each column uses the step size selected by Hit@1 from the common sweep in App. B.1; Hit@5 and Hit@10 are evaluated at the same selected value.

Metric	$K=1$	$K=4$	$K=16$	$K=64$	$K=256$
Hit@1	44.0	38.5	37.7	33.4	35.4
Hit@5	75.2	70.1	70.5	69.2	71.5
Hit@10	82.9	80.3	81.4	80.5	81.7
Selected α	3	4	8	8	3

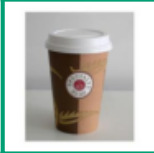
Reference	Text Edit	Rank 1	Rank 2	Rank 3
	The cup is brown, nor red. target at rank 1			
	Four empanadas on a plate. target at rank 1			
	Ox hangs out by the water. target at rank 1			
	Different patterns on the towel, only two colours instead of four. target at rank 1			
	Make a person holding the crab. target at rank 1			
	Show multiple ostriches. target at rank 2			
	Two dogs of the same breed on the floor. target not in top 5			

Figure 23 Composed queries and the top-3 retrieval results. Green borders indicate the ground-truth targets. While the overall retrieval results reflect that editing instructions correctly, the final two rows also highlight a zero-shot limitation: simultaneously preserving the reference category while applying count edits remains challenging.

H Ablation on the training loss contributions

FLAT pre-training incorporates three loss terms targeting retrieval, T2I generation, and I2T generation. We perform an ablation study by training models on different subsets of these losses to evaluate their individual and joint contributions. Table 15 reports downstream scores across all evaluated prefix lengths and loss combinations. Each variant uses the same initialization and 40k-step schedule, varying only the active loss components. Incorporating all three losses provides a clear advantage, yielding balanced, optimal performance across all metrics and prefix lengths K . Figure 24 plots the normalized evaluation scores for each ablated variant across all prefix lengths.

Table 15 Training loss ablation using all prefix lengths. $\mathcal{L}_{\text{ctr}}$, $\mathcal{L}_{\text{cap}}$, and $\mathcal{L}_{\text{gen}}$ denote contrastive, captioning, and image-generation loss respectively.

$K=1$						
Objectives	I2T	T2I	B@4	CIDEr	Gen.	
$\emptyset$	0.1	0.2	24.4	80.8	0.000	
$\mathcal{L}_{\text{ctr}}$	50.7	57.2	33.9	117.3	0.001	
$\mathcal{L}_{\text{cap}}$	48.2	46.3	35.9	123.3	0.004	
$\mathcal{L}_{\text{gen}}$	41.7	49.6	32.1	106.6	0.328	
$\mathcal{L}_{\text{ctr}} + \mathcal{L}_{\text{cap}}$	56.1	59.2	35.3	122.2	0.004	
$\mathcal{L}_{\text{ctr}} + \mathcal{L}_{\text{gen}}$	52.6	54.1	33.8	117.2	0.280	
$\mathcal{L}_{\text{cap}} + \mathcal{L}_{\text{gen}}$	54.2	55.4	36.1	123.9	0.298	
All three	57.8	62.4	35.4	122.7	0.329	

$K=4$						
Objectives	I2T	T2I	B@4	CIDEr	Gen.	
$\emptyset$	0.1	0.0	27.3	91.5	0.000	
$\mathcal{L}_{\text{ctr}}$	50.8	57.0	34.1	118.1	0.002	
$\mathcal{L}_{\text{cap}}$	26.4	21.2	38.7	133.7	0.005	
$\mathcal{L}_{\text{gen}}$	41.8	50.1	32.1	107.0	0.323	
$\mathcal{L}_{\text{ctr}} + \mathcal{L}_{\text{cap}}$	55.1	56.2	38.3	131.9	0.005	
$\mathcal{L}_{\text{ctr}} + \mathcal{L}_{\text{gen}}$	52.4	54.7	34.1	117.4	0.309	
$\mathcal{L}_{\text{cap}} + \mathcal{L}_{\text{gen}}$	36.2	36.6	38.6	133.1	0.298	
All three	53.9	60.1	37.6	130.4	0.325	

$K=16$						
Objectives	I2T	T2I	B@4	CIDEr	Gen.	
$\emptyset$	0.2	0.0	30.6	104.3	0.000	
$\mathcal{L}_{\text{ctr}}$	50.4	56.7	34.1	117.7	0.002	
$\mathcal{L}_{\text{cap}}$	28.0	19.8	40.3	138.3	0.004	
$\mathcal{L}_{\text{gen}}$	10.1	40.6	32.9	110.4	0.325	
$\mathcal{L}_{\text{ctr}} + \mathcal{L}_{\text{cap}}$	53.5	56.0	40.0	137.2	0.003	
$\mathcal{L}_{\text{ctr}} + \mathcal{L}_{\text{gen}}$	51.5	53.4	33.9	117.0	0.328	
$\mathcal{L}_{\text{cap}} + \mathcal{L}_{\text{gen}}$	41.6	41.0	40.1	137.8	0.294	
All three	54.3	58.4	39.7	136.8	0.324	

$K=64$						
Objectives	I2T	T2I	B@4	CIDEr	Gen.	
$\emptyset$	0.2	0.0	31.0	105.3	0.000	
$\mathcal{L}_{\text{ctr}}$	50.4	56.9	34.0	117.7	0.003	
$\mathcal{L}_{\text{cap}}$	35.6	21.0	40.7	139.3	0.004	
$\mathcal{L}_{\text{gen}}$	1.1	4.4	33.0	110.4	0.327	
$\mathcal{L}_{\text{ctr}} + \mathcal{L}_{\text{cap}}$	52.7	54.8	39.9	137.8	0.004	
$\mathcal{L}_{\text{ctr}} + \mathcal{L}_{\text{gen}}$	51.7	53.7	33.8	116.8	0.299	
$\mathcal{L}_{\text{cap}} + \mathcal{L}_{\text{gen}}$	42.7	37.3	40.5	138.7	0.296	
All three	52.9	56.7	40.1	137.5	0.319	

$K=256$						
Objectives	I2T	T2I	B@4	CIDEr	Gen.	
$\emptyset$	0.2	0.0	30.8	105.0	0.000	
$\mathcal{L}_{\text{ctr}}$	51.0	57.0	34.2	118.0	0.003	
$\mathcal{L}_{\text{cap}}$	39.4	4.0	40.8	139.7	0.004	
$\mathcal{L}_{\text{gen}}$	0.6	1.1	33.1	110.8	0.323	
$\mathcal{L}_{\text{ctr}} + \mathcal{L}_{\text{cap}}$	51.2	54.5	40.0	137.8	0.003	
$\mathcal{L}_{\text{ctr}} + \mathcal{L}_{\text{gen}}$	51.5	53.6	33.9	117.1	0.329	
$\mathcal{L}_{\text{cap}} + \mathcal{L}_{\text{gen}}$	13.6	6.7	40.7	139.2	0.297	
All three	52.2	56.3	40.2	137.5	0.327	

We also compute Shapley values (Shapley, 1953) to quantify the marginal contribution of each loss objective. For a loss component i and downstream metric v , the Shapley attribution is defined as:

$$\phi_i(v) = \sum_{S \subseteq \mathcal{O} \setminus i} \frac{|S|!(|\mathcal{O}| - |S| - 1)!}{|\mathcal{O}|!} [v(S \cup i) - v(S)], \quad (10)$$

where $\mathcal{O}$ is the set of all three losses. Figure 25 visualizes these Shapley values across prefix lengths. The heatmaps normalize each attribution by $v(\mathcal{O}) - v(\emptyset)$, ensuring that each metric column sums to 100%. The task-aligned attribution structure becomes sharper as the prefix widens: contrastive learning increasingly accounts for retrieval performance, captioning dominates both text-generation metrics from $K=16$ onward, and image generation consistently accounts for at least 95% of GenEval. Minor negative entries at $K=16$ and $K=256$ indicate that auxiliary loss contributions are not uniformly positive across all prefix lengths, although the primary task-aligned attributions remain dominant.

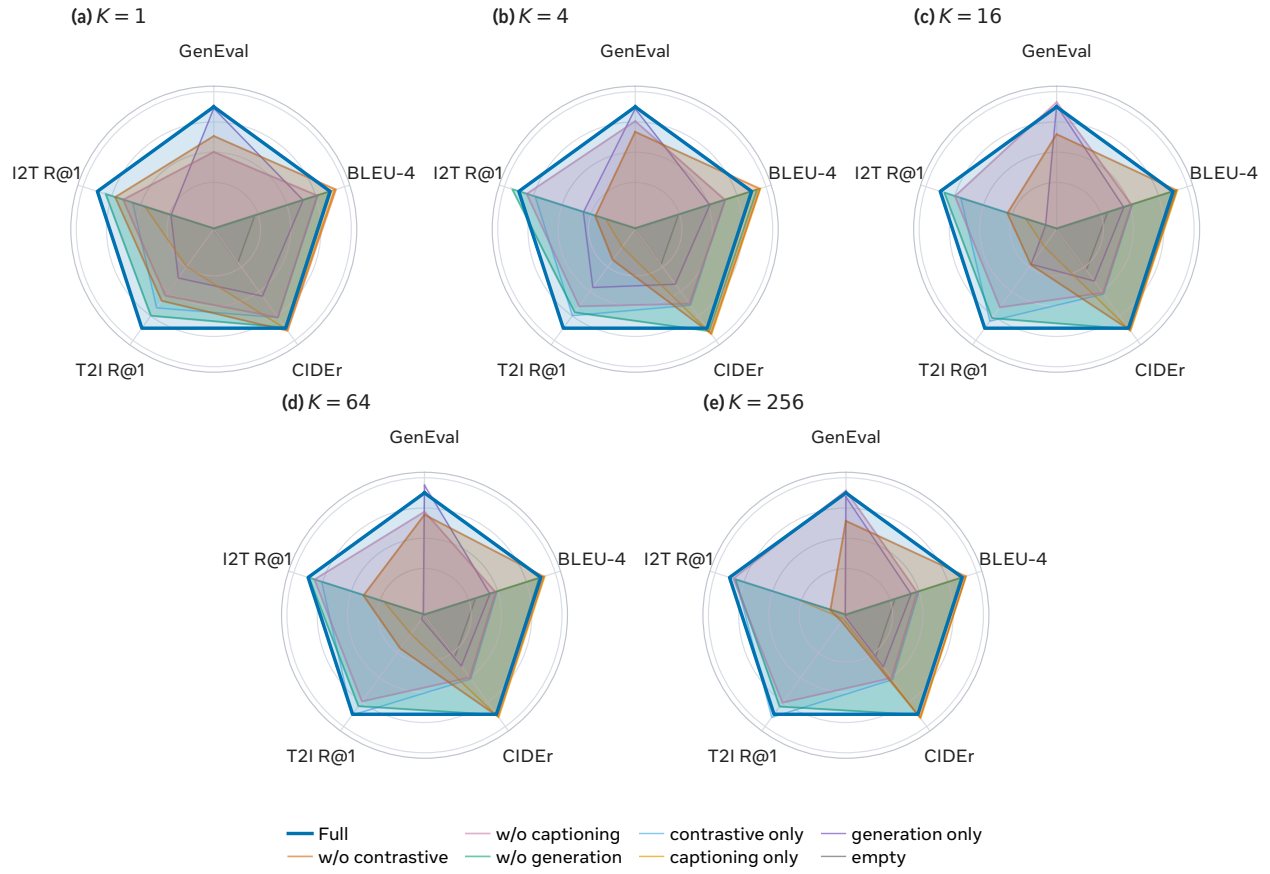


Figure 24 Ablation on training loss. Eval performance at $K \in \{1, 4, 16, 64, 256\}$. Each axis is normalized at the same prefix length for plot.



Figure 25 Relative Shapley attribution at different prefix lengths. Cell values are percentages of the total improvement over the untrained checkpoint for each metric. Blue denotes positive attribution and orange denotes negative attribution.

I Truncated prefixes and null padding

Section 3.1 describes truncating the register sequence to its first K positions. An alternative strategy retains a fixed sequence length N and replaces positions $i \geq K$ with learned null embeddings, as in FlexTok (Bachmann et al., 2025). Empirically, we find that null-padded token sequences introduce optimization instabilities during FLAT joint training. Figure 26 illustrates these training dynamics. Under null padding, the gradient norm grows rapidly and spikes near 33k steps, causing the contrastive loss to collapse toward $\ln(1024)$ —the theoretical random-baseline value for the global batch. In contrast, prefix truncation remains stable throughout the same training window. As shown in Figure 27, null-padded conditioning loses semantic correspondence between the source image and its generated counterpart, whereas prefix truncation preserves recognizable semantic relationships. Because these runs follow their respective original training recipes (including differing K -sampling granularities and learning-rate schedules), we interpret these findings as qualitative evidence of training dynamics.

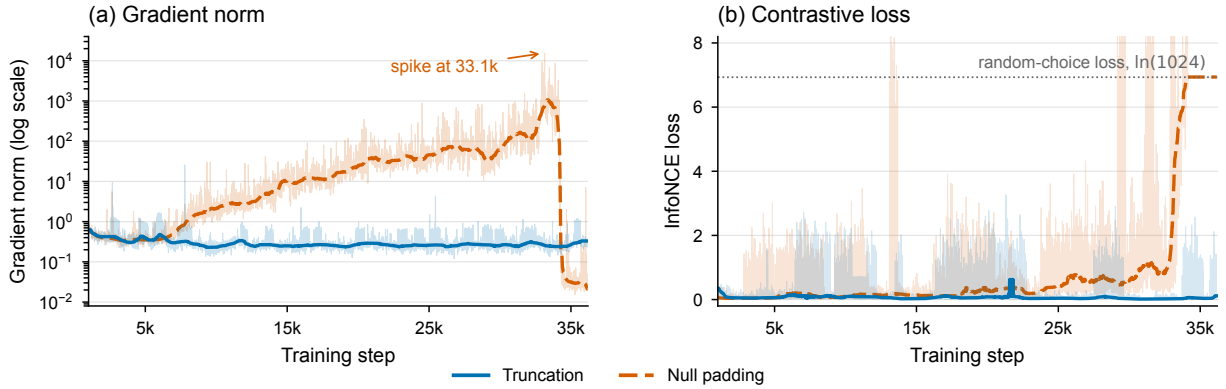


Figure 26 Optimization behavior under truncation and null padding. Curves are smoothed with a 1,010-step rolling median; faint lines show the unsmoothed measurements. The null-padding run develops large gradient excursions before its contrastive loss approaches the random-choice value.



Figure 27 Images generated by prefix truncation v.s. null padding at step 40k. Each column uses the same COCO target-caption pair and diffusion noise.